



NVIDIA Nemotron

Open Models, Datasets, and Techniques for Agentic AI

NVIDIA Leads Open-Source AI Contributions



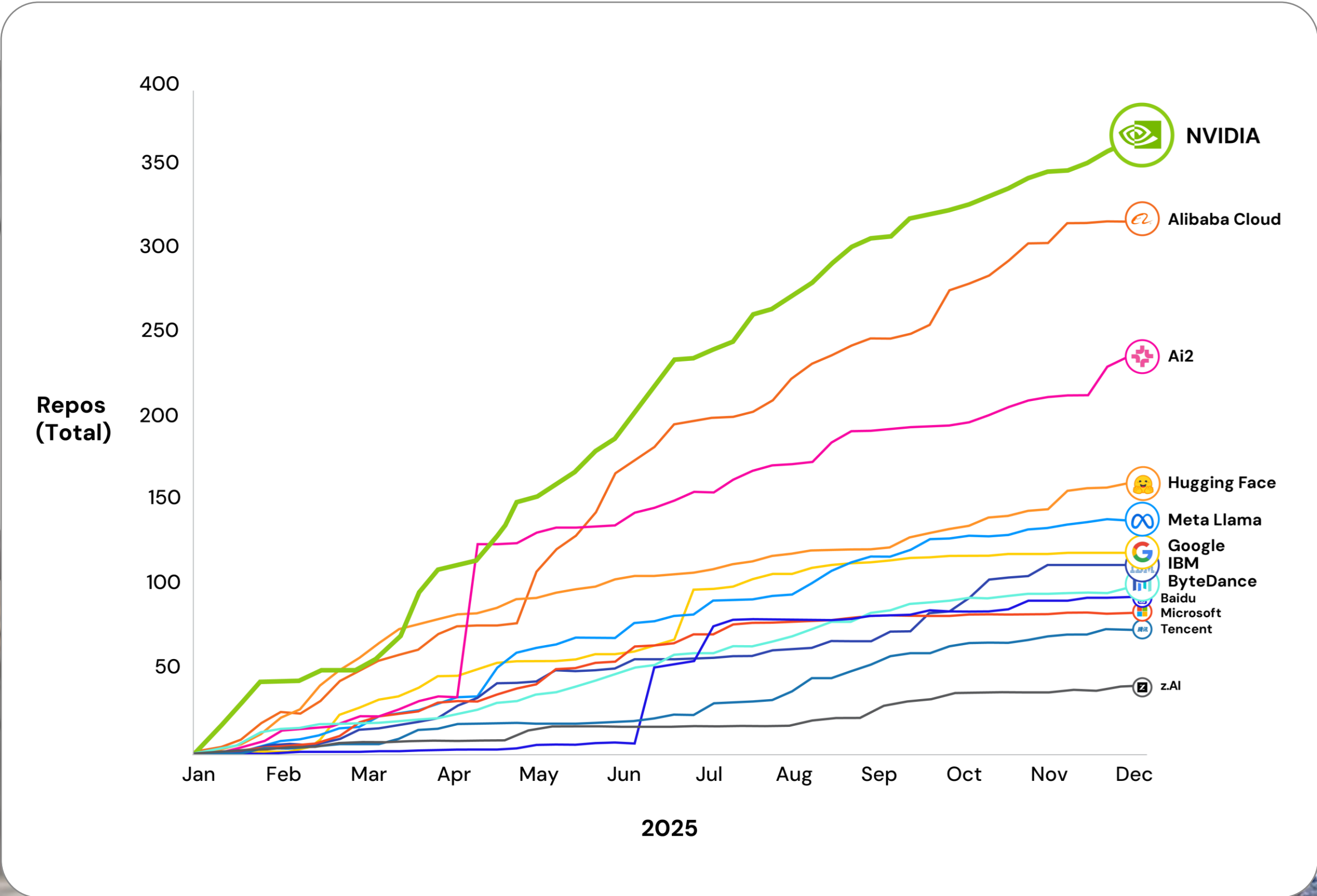
Clara
Biomedical AI



Earth-2
AI-Physics



Nemotron
Agentic AI



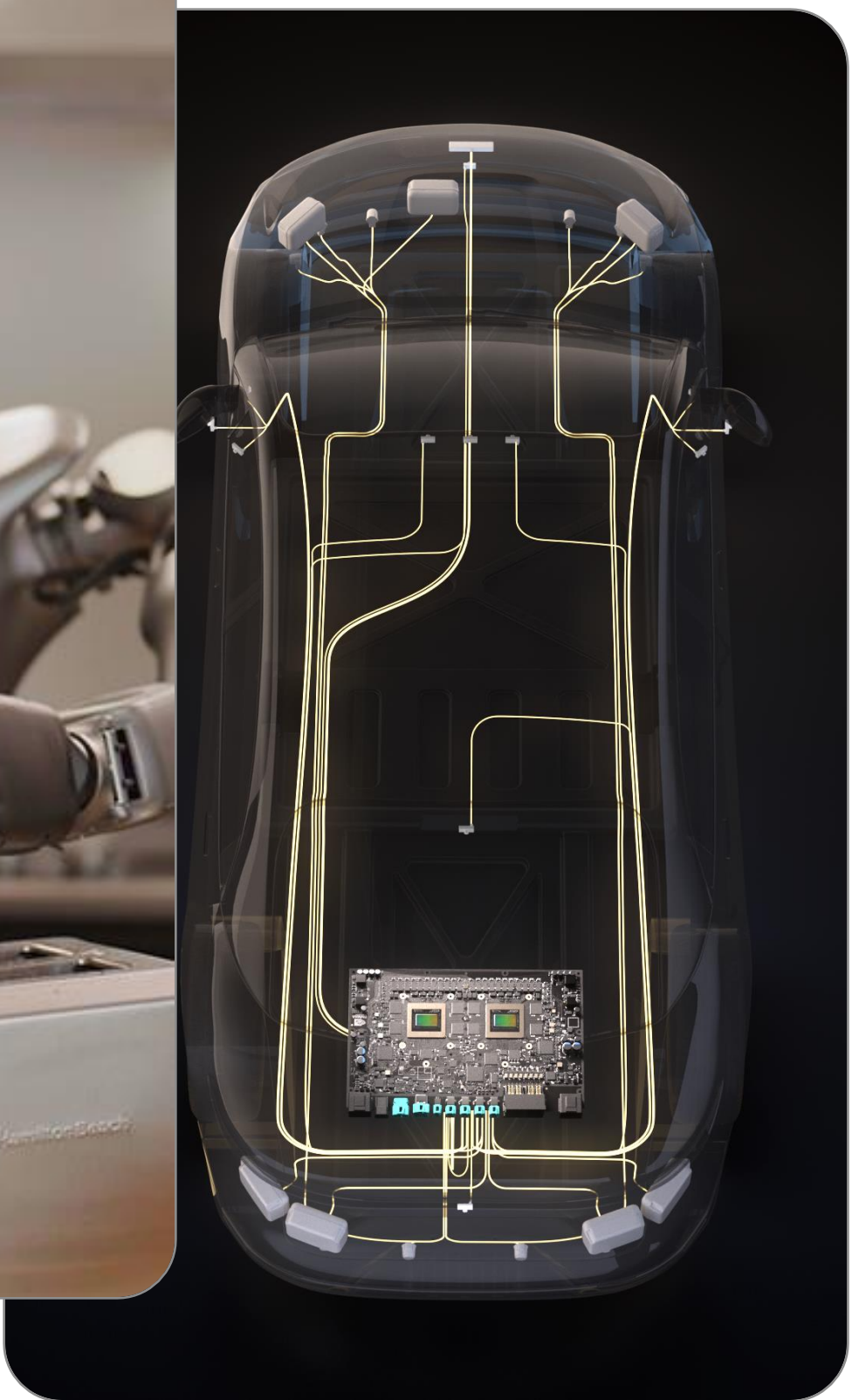
NVIDIA Leads AI Open Source
With Repository Growth



Cosmos
Physical AI



GROOT
Robotics



Alpamayo
Autonomous Vehicle

The Evolution of AI Agents

From LLMs and multi-turn prompts to long-running agents that complete tasks

ChatGPT

Nov 2022



Foundation LLMs,
Multi-turn Prompts

DeepSeek

Jan 2025

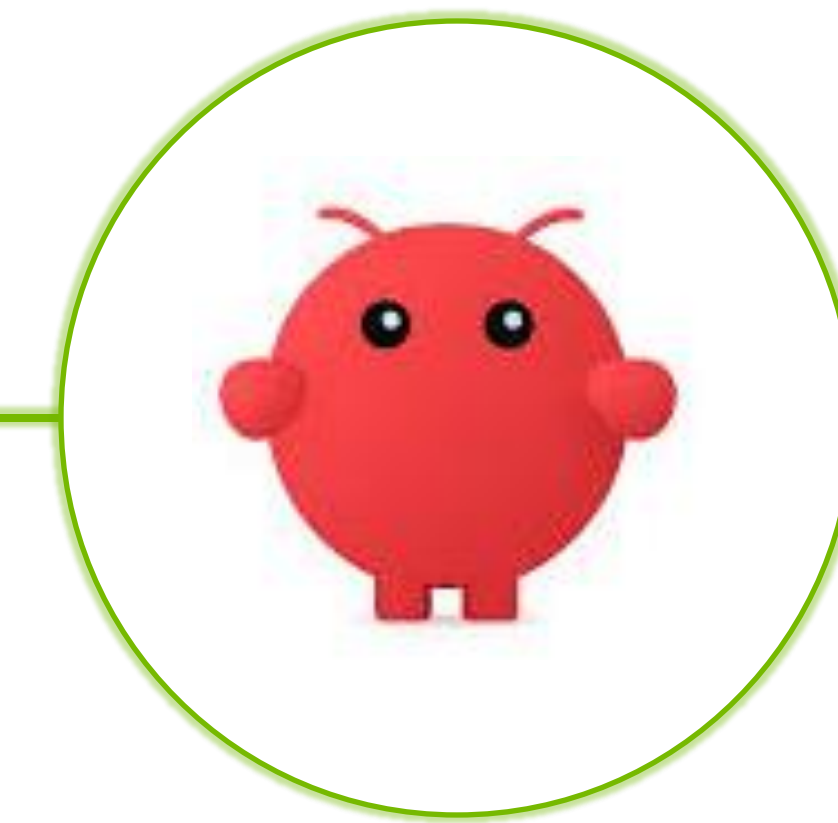


Reasoning Enables Reflection
For Better Outcomes

100x

OpenClaw

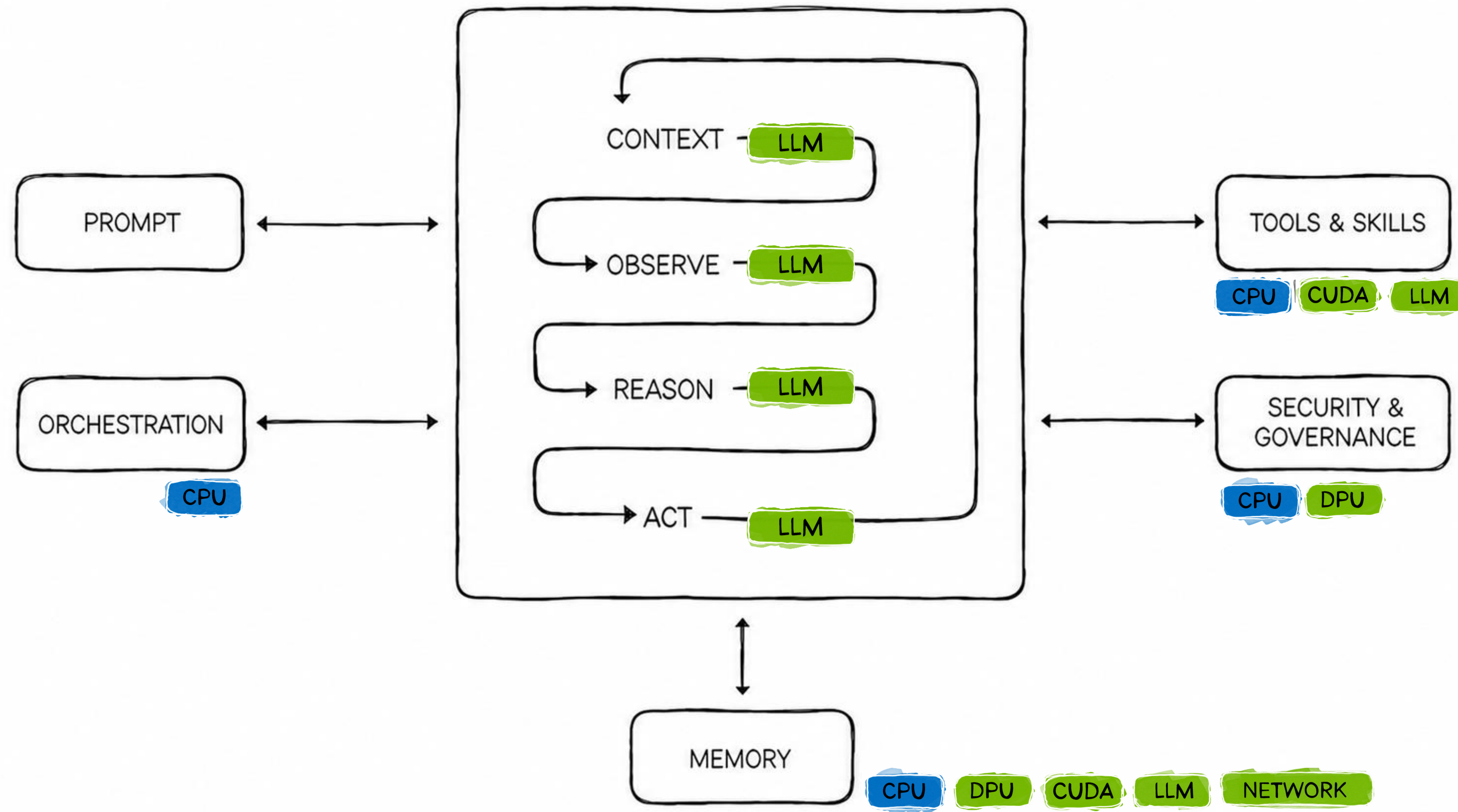
Jan 2026



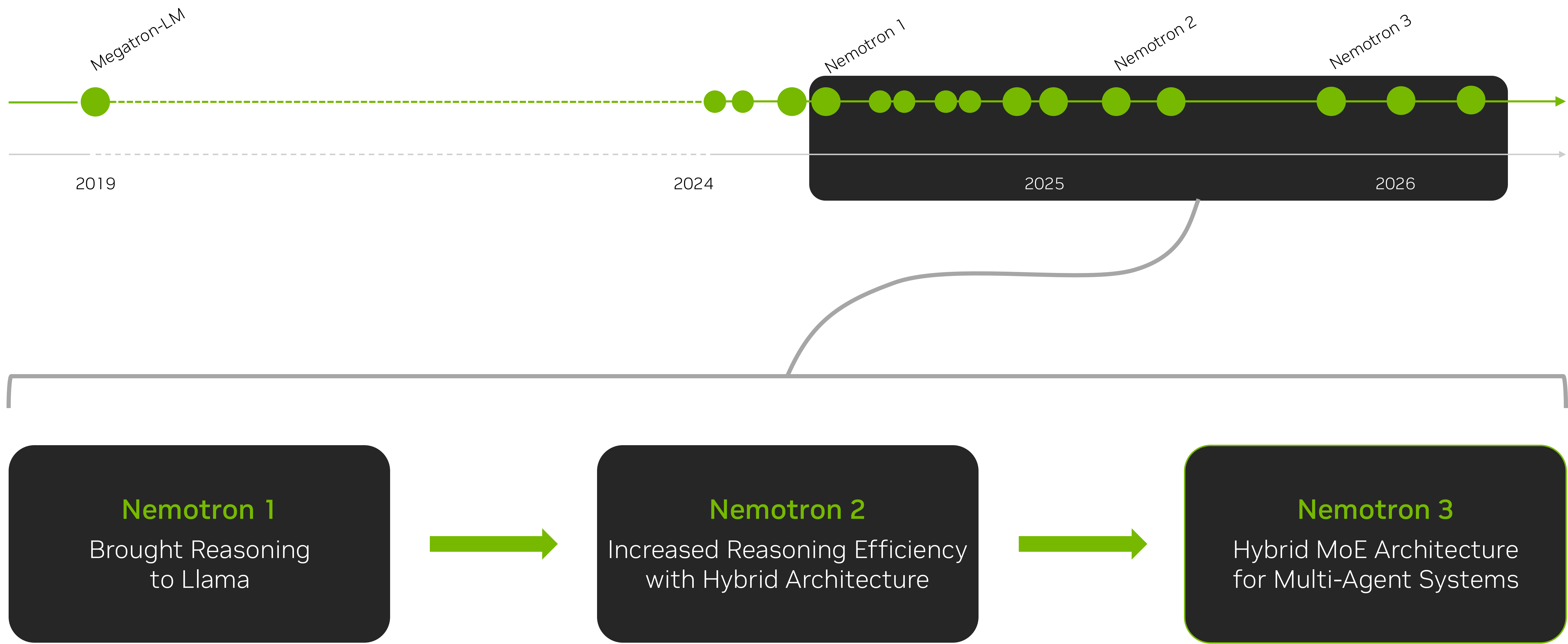
Self Evolving Agents
Working on Your Behalf

1000x

How Agents Work



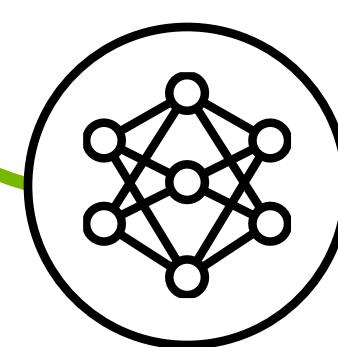
Commitment to Open Models



NVIDIA Nemotron Open Models, Data, Libraries for Agentic AI

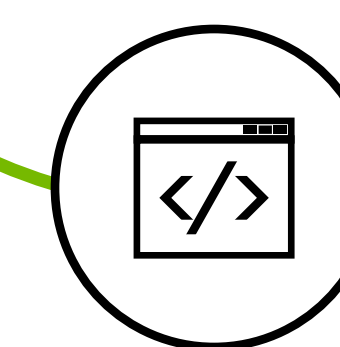


Nemotron 3
Nemotron VL
Nemotron Speech
Nemotron RAG
Nemotron Safety



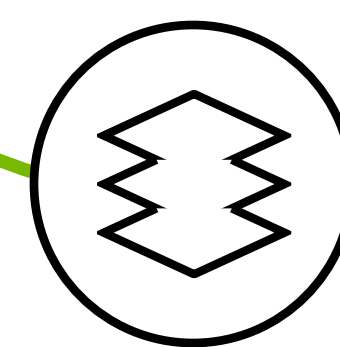
Models

Nemotron Cookbook
Nemotron Recipes



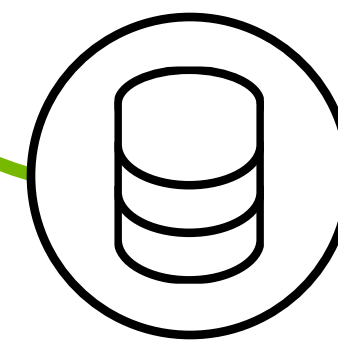
Training Scripts

NeMo Data
Megatron Core
NeMo RL & Gym
NeMo Evaluator
NeMo Agent Toolkit



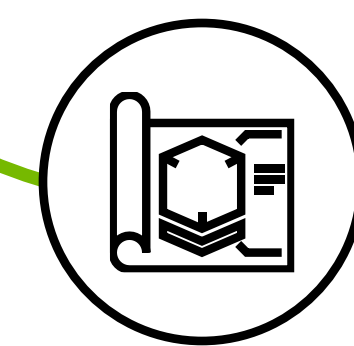
Libraries

10 Trillion Pretraining Tokens
40 Million Post-training Samples
900K RL Tasks
11K Safety Traces



Data

Deep Research
Video Search and Summarization
RAG
LLM Router



Blueprints

NVIDIA Nemotron Models Available Today

Reasoning

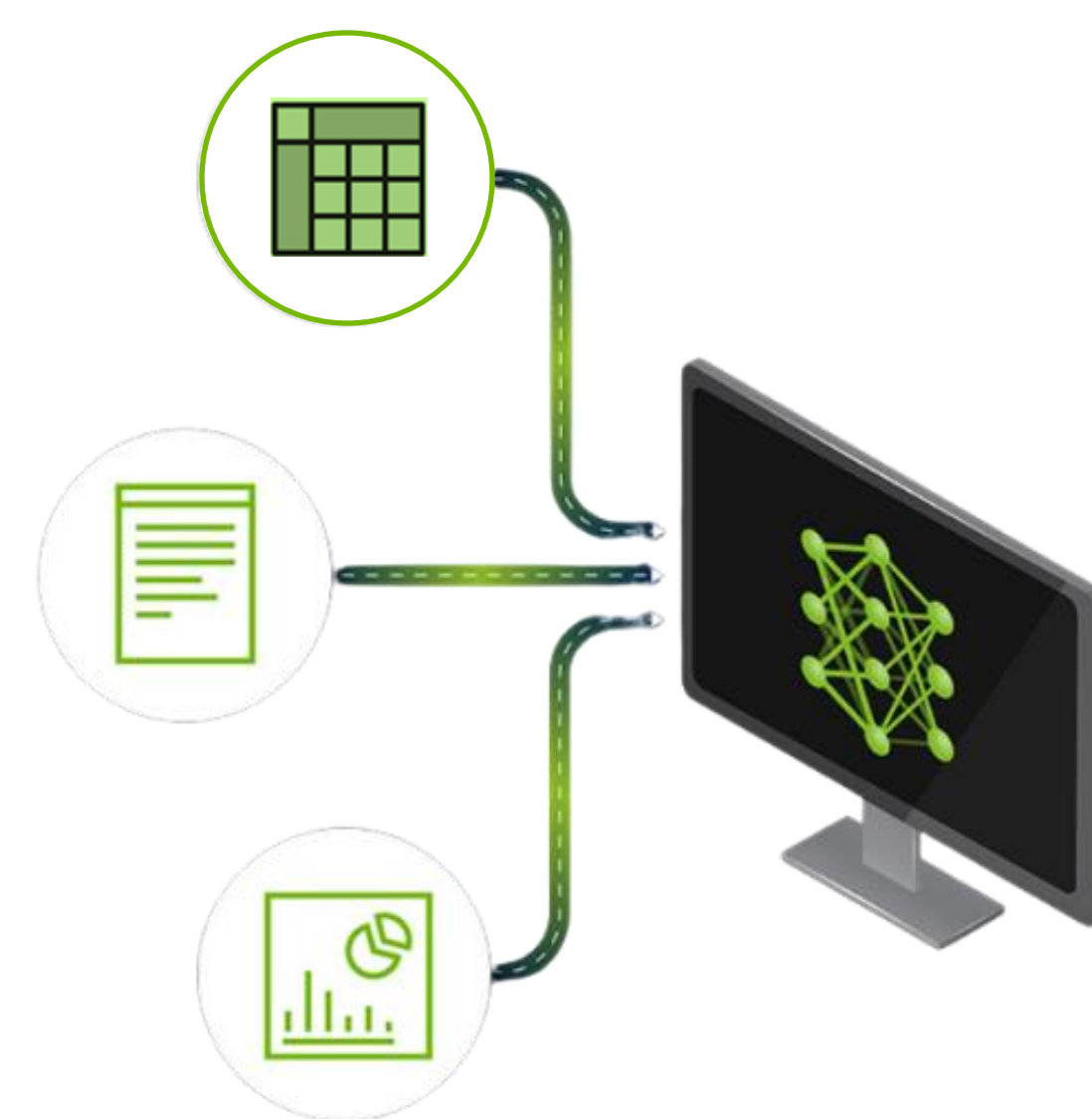


Nemotron 3 Nano
Highest efficiency SLM

Nemotron 3 Super
Highest efficiency LLM

Nemotron 3 Ultra
Frontier-reasoning and orchestration

Vision

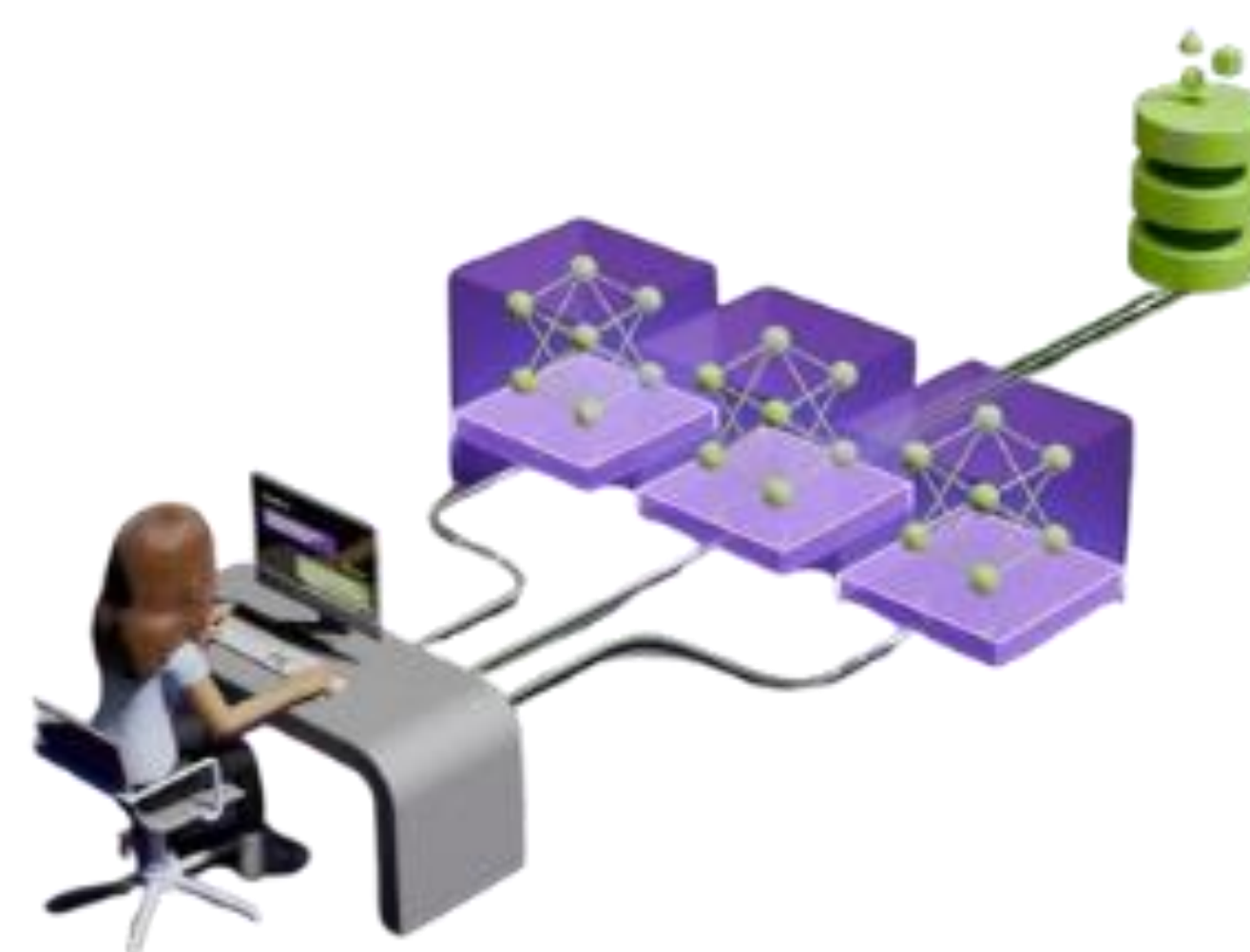


Nemotron Nano 2 VL
Video understanding and Image reasoning

Nemotron Parse 1.1
Document extraction

Nemotron 3 Nano Omni
Enterprise-grade multimodal understanding

Information Retrieval



Nemotron Page Elements
Structured images detection & classification

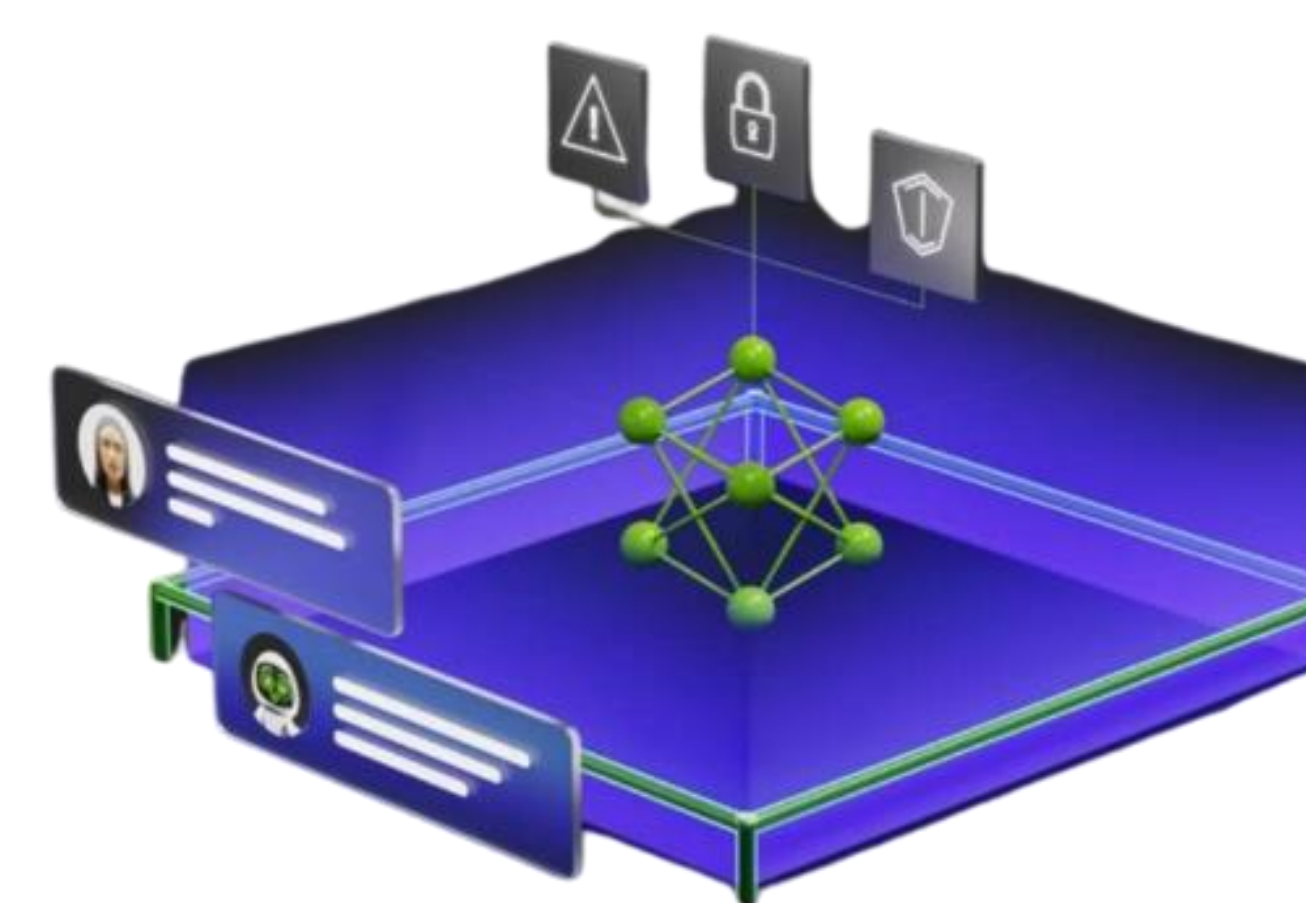
Nemotron Table Structure
Rows, columns detection & preservation in markdown

Nemotron Graphic Elements
Chart component detection & classification

Llama Nemotron Embed
Multilingual text question-answering retrieval

Llama Nemotron Rerank
Multilingual fine-tuned reranking model

Content Safety



Llama Nemotron Safety Guard
Multilingual content safety

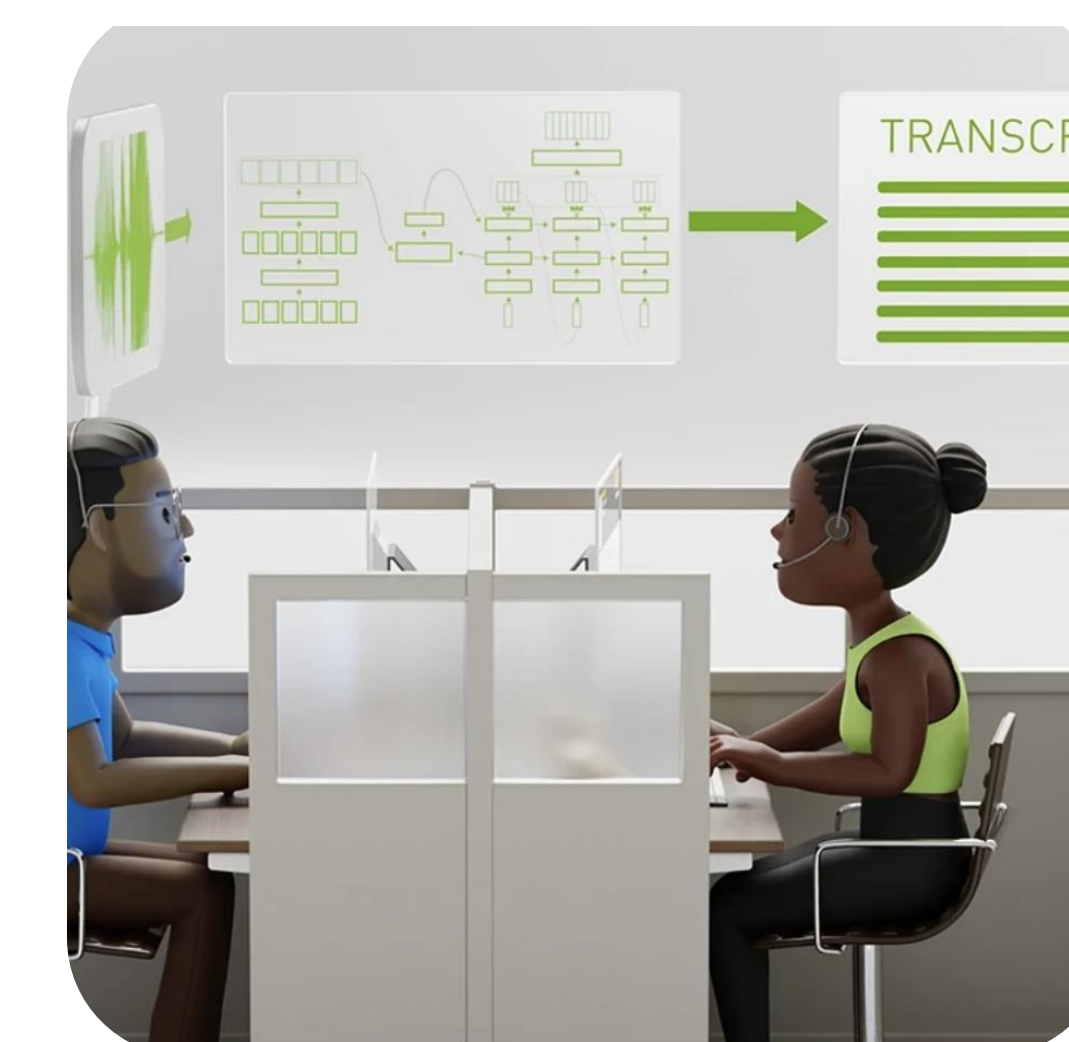
Llama Nemotron Content Safety
Multilingual content safety

Nemotron Content Safety Reasoning
Domain-specific safety policies

Llama Nemotron Topic Control
Prevent off-topic / unsafe deviations

Nemotron Jailbreak Detection
Detect adversarial or jailbreak prompts to block unsafe behavior

Speech



Nemotron ASR
Real-time English speech recognition

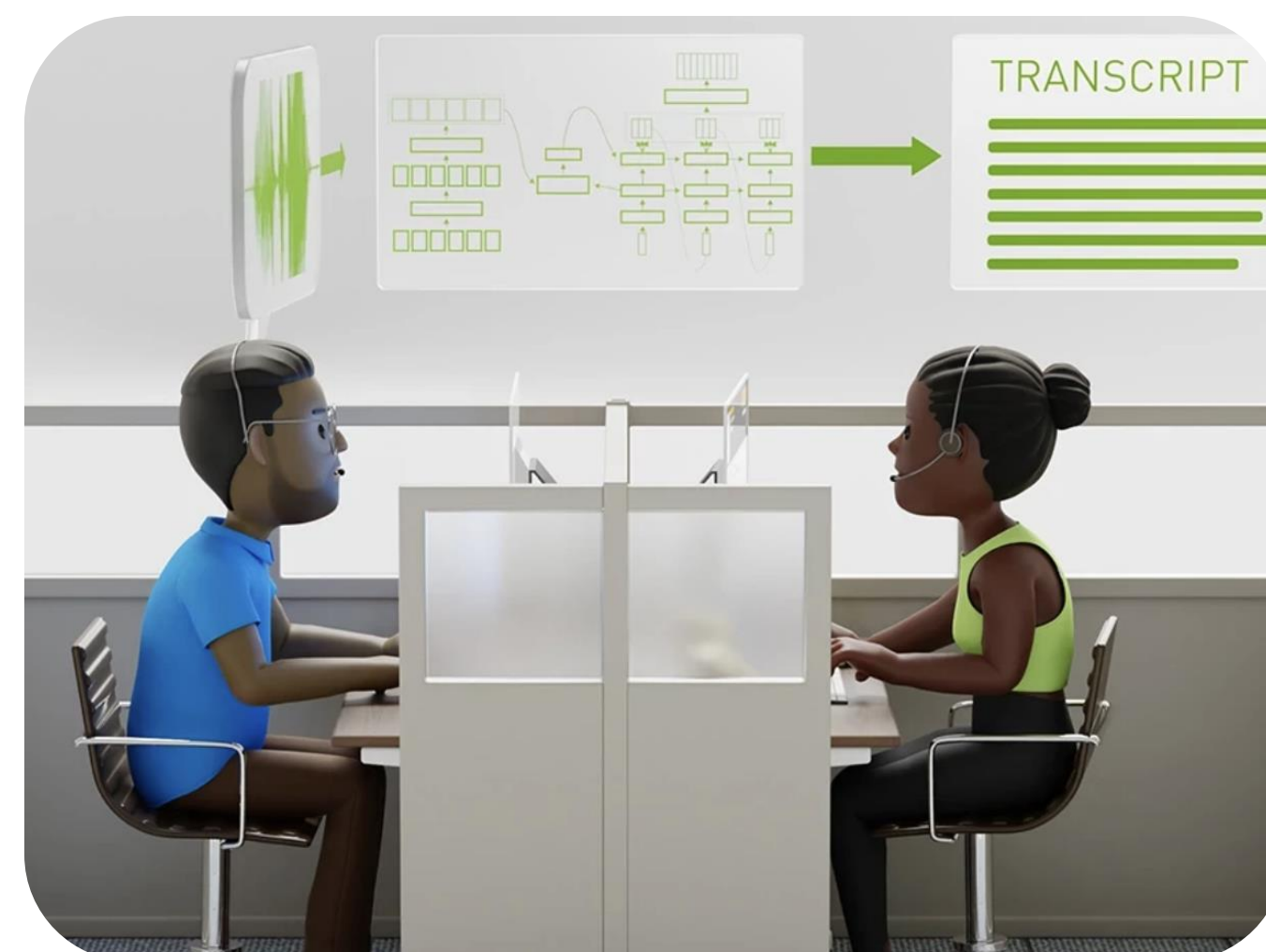
Magpie TTS Multilingual
Research-ready, text-to-speech for natural, fine-tunable voices across 9 languages

Nemotron 3 VoiceChat
Full duplex speech-to-speech 12B model built for voice agents
(Early Access)

NVIDIA Nemotron Speech

Open, production-ready speech models with leaderboard-topping ASR, TTS, NMT, and speech-to-speech.

Automatic Speech Recognition (ASR)



Nemotron 3.5 ASR:
~40 languages

Top 15 fastest models on
Hugging Face Open ASR
Leaderboard (up to 6,399x
real-time)

Streaming, batch, and hybrid
ASR-LLM modes across the
model family including
Parakeet and Canary

Text to Speech (TTS)



Expressive audio from text in 9
languages from Magpie TTS

Lowest CER/WER and highest
naturalness (MOS) among open-
source TTS models

Voice cloning from a 5-sec
sample; sub-200ms latency

Neural Machine Translation (NMT)



Any-to-any translation across 36
languages

Term protection via do-not-
translate tags for brand names
and domain terms

Pairs with ASR + TTS for end-to-
end speech-to-speech translation
pipelines

Speech to Speech



Unified ASR + LLM + TTS in one
architecture; English

Replaces the cascaded stack

Sub-second latency with barge-
in, backchannels, and
overlapping speech

Built on Nemotron Nano v2
backbone

[Early Access](#)



Nemotron Speech Momentum and Market Signal

The Developer Community: Impact of Hugging Face ASR Leaderboard Wins



The 🗨️ Open ASR Leaderboard ranks and evaluates speech recognition models on the Hugging Face Hub. We report the Average [WER](#) (📉 lower the better) and [RTFx](#) (📈 higher the better). Models are ranked based on their Average WER, from lowest to highest. Check the [Metrics](#) tab to understand how the models are evaluated. If you want results for a model that is not listed here, you can submit a request for it to be included ✨. The leaderboard includes both English ASR evaluation and multilingual benchmarks across the top European languages.

model	Average WER 📉	RTFx 📈	License	AMI	Earnings22	Gigaspeech	LS Clean	LS Other	SPGISpeech	Tedlium	Vox
ibm-granite/granite-4.0-1b-speech	5.52	280.02	Open	8.44	8.48	10.14	1.42	2.85	3.89	3.1	5.8
nvidia/canary-qwen-2.5b	5.63	418.28	Open	10.19	10.45	9.43	1.61	3.1	1.9	2.71	5.6
ibm-granite/granite-speech-3.3-8b	5.74	145.42	Open	8.98	9.42	10.19	1.43	2.86	3.91	3.4	5.7
Qwen/Qwen3-ASR-1.7B	5.76	147.93	Open	10.56	10.25	8.74	1.63	3.4	2.84	2.28	6.3
elevenlabs/scribe_v2	5.83	NA	Proprietary	11.86	9.43	9.11	1.54	2.83	2.68	2.37	6.8
ibm-granite/granite-speech-3.3-2b	6	270.57	Open	8.9	10.25	10.69	1.53	3.26	3.87	3.57	5.9
microsoft/Phi-4-multimodal-instruct	6.02	151.1	Open	11.09	10.16	9.33	1.69	3.82	3.06	2.94	6.0
nvidia/parakeet-tdt-0.6b-v2	6.05	3386.02	Open	11.16	11.15	9.74	1.69	3.19	2.17	3.38	5.9
assemblyai/universal-3-pro	6.21	NA	Proprietary	13.85	10.9	9.14	1.53	2.7	1.68	3.01	6.8
aquavoice/avalon-v1-en	6.24	NA	Proprietary	11.58	11.37	9.52	1.68	3.28	2.1	3.02	7.3
nvidia/parakeet-tdt-0.6b-v3	6.32	3332.74	Open	11.39	11.19	9.57	1.92	3.59	3.98	2.8	6.0
nvidia/canary-1b-flash	6.35	1045.75	Open	13.11	12.77	9.85	1.48	2.87	1.95	3.12	5.0

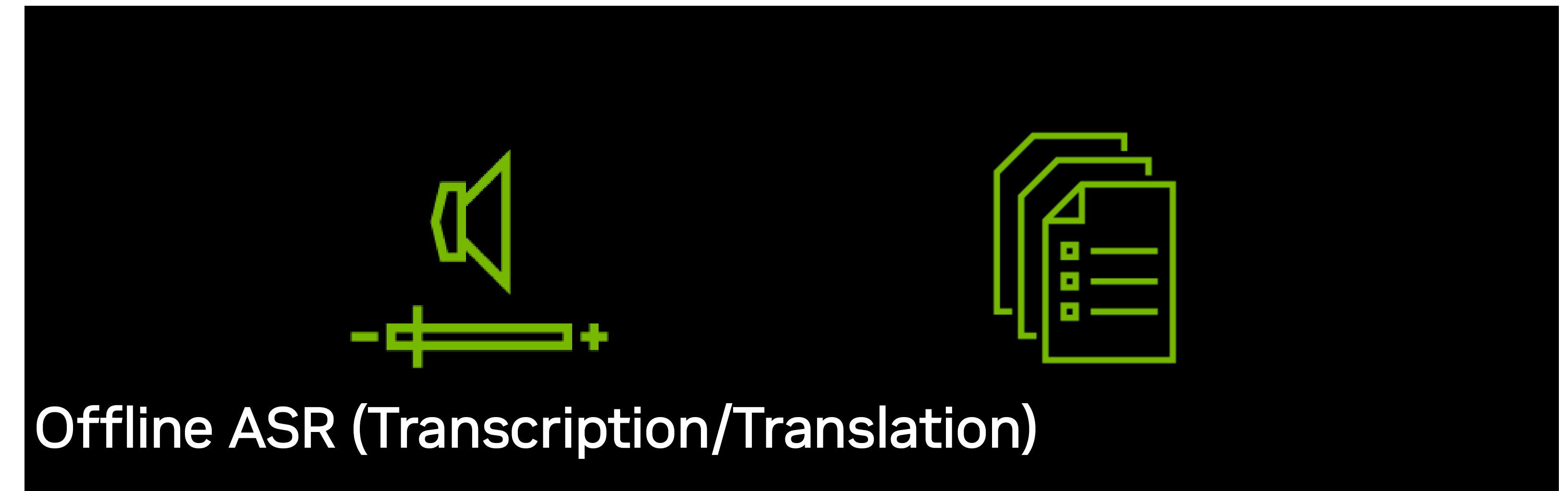
model	Average WER 📉	RTFx... 📈
nvidia/parakeet-tdt_ctc-110m	7.49	5345.14
nvidia/parakeet-ctc-0.6b	7.69	4281.53
nvidia/parakeet-tdt-0.6b-v2	6.05	3386.02
nvidia/parakeet-tdt-0.6b-v3	6.32	3332.74
nvidia/parakeet-rnnt-0.6b	7.5	2815.72
nvidia/parakeet-ctc-1.1b	7.4	2728.52
nvidia/parakeet-tdt-1.1b	7.01	2390.61
nvidia/parakeet-rnnt-1.1b	7.12	2053.15
nvidia/canary-180m-flash	7.12	1233.58
nvidia/canary-1b-flash	6.35	1045.75
nvidia/canary-1b-v2	7.15	749

Streaming vs Offline/Batch ASR

Speech AI for Low-Latency and High-Fidelity Workflows



- Real-time transcription for low latency & high accuracy in noisy environments
- Primarily used in live voice assistants, meetings, and customer service applications
- **CTC and RNNT architectures support streaming**
- Key Models:
 - Nemotron Speech ASR (English)
 - Parakeet-ctc-1.1b
 - Parakeet-ctc-0.6b (Spanish, Tw-CS, Vi-CS, Zh-CS)
 - Parakeet-RNNT-Multilingual (default, prompt, Indic)



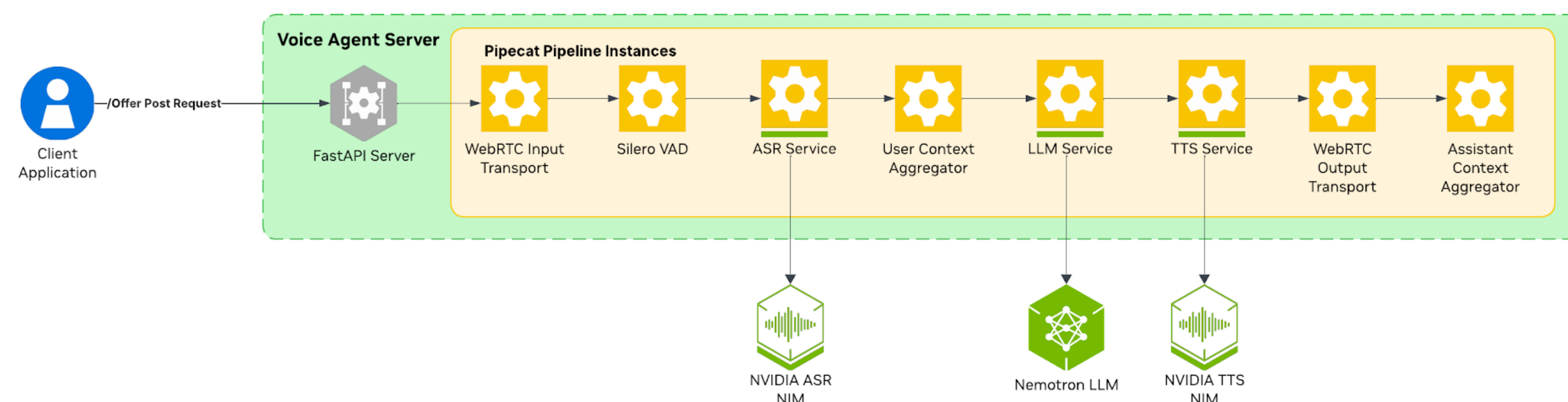
- File-based, batch transcription/translation for timestamped and multi-language audio
- Used for processing pre-recorded calls, videos, or archival data
- **TDT and Canary architectures are offline**
- Key Models:
 - Parakeet-tdt-0.6b-v2/v3
 - Canary-1b-Flash
 - Whisper-Large-v3
- Supports transcription, translation, summarization, and multilingual inputs

Choosing the Right Speech Workflow

Cascaded vs Speech to Speech

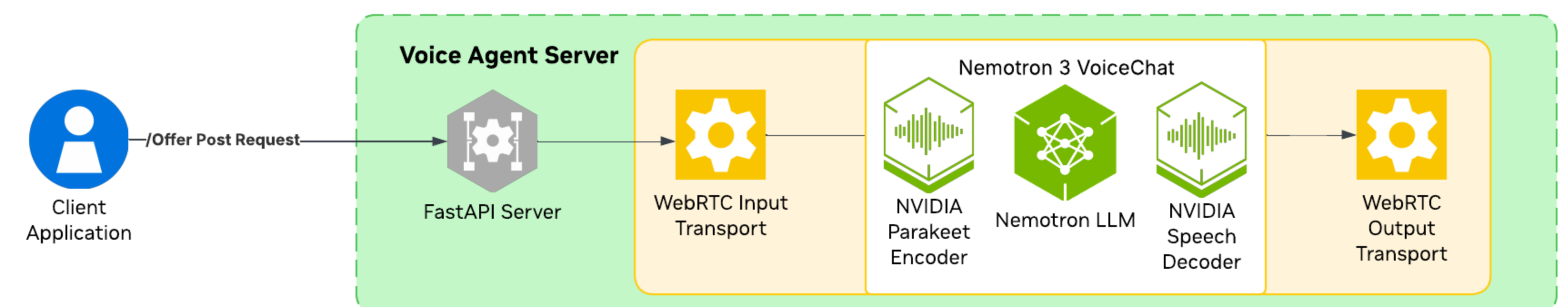
Classic Cascaded Voice Pipeline

- ASR → LLM/agent → TTS
- Best when:
- You need **maximum control** over each stage
- Tool-calling, RAG, and complex workflows already exist
- You want to **swap or tune** ASR, LLM, or TTS independently
- Multilingual use cases



End-to-End Speech-to-Speech

- Single full-duplex S2S model (listens and speaks at once)
- Best when:
- **Natural conversation and latency** are the top priorities
- You want **barge-in**, backchannels, overlapping speech along with **backchanneling** (eg: "mm-hm", "right"...)
- You're building new voice agents where a unified model simplifies ops





NVIDIA Nemotron Benchmarks

Nemotron 3 Ultra: Smaller, Smarter Frontier-Intelligence Model

	Nemotron 3 Ultra (550B)	GLM 5.1 (744B)	Kimi K2.6 (1T)	Qwen3.5 (397B)
Agent Productivity PinchBench	91%	84%	91%	89%
Long-horizon Planning EnterpriseOps-Gym	33%	40%	29%	30%
Coding Terminal-Bench 2.0	54%	64%	67%	53%
Instruction Following IFBench	82%	77%	74%	78%
Knowledge Work GDPVal-AA	1,448	1,594	1,508	1,192
Professional Work Tasks ProfBench (Search)	56%	46%	56%	53%
Long Context Ruler @1M	95%	N/A (max 256K)	N/A (max 256K)	90%

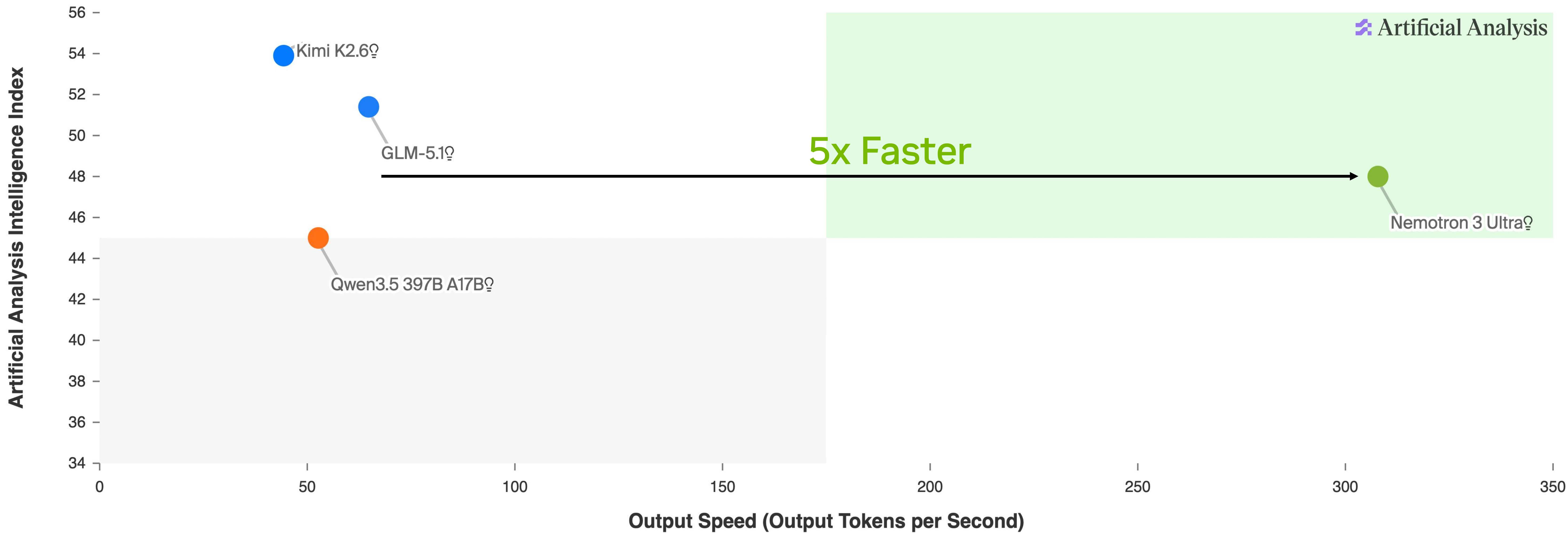
Nemotron 3 Ultra Delivers 5x Faster Inference

Intelligence vs. Output Speed

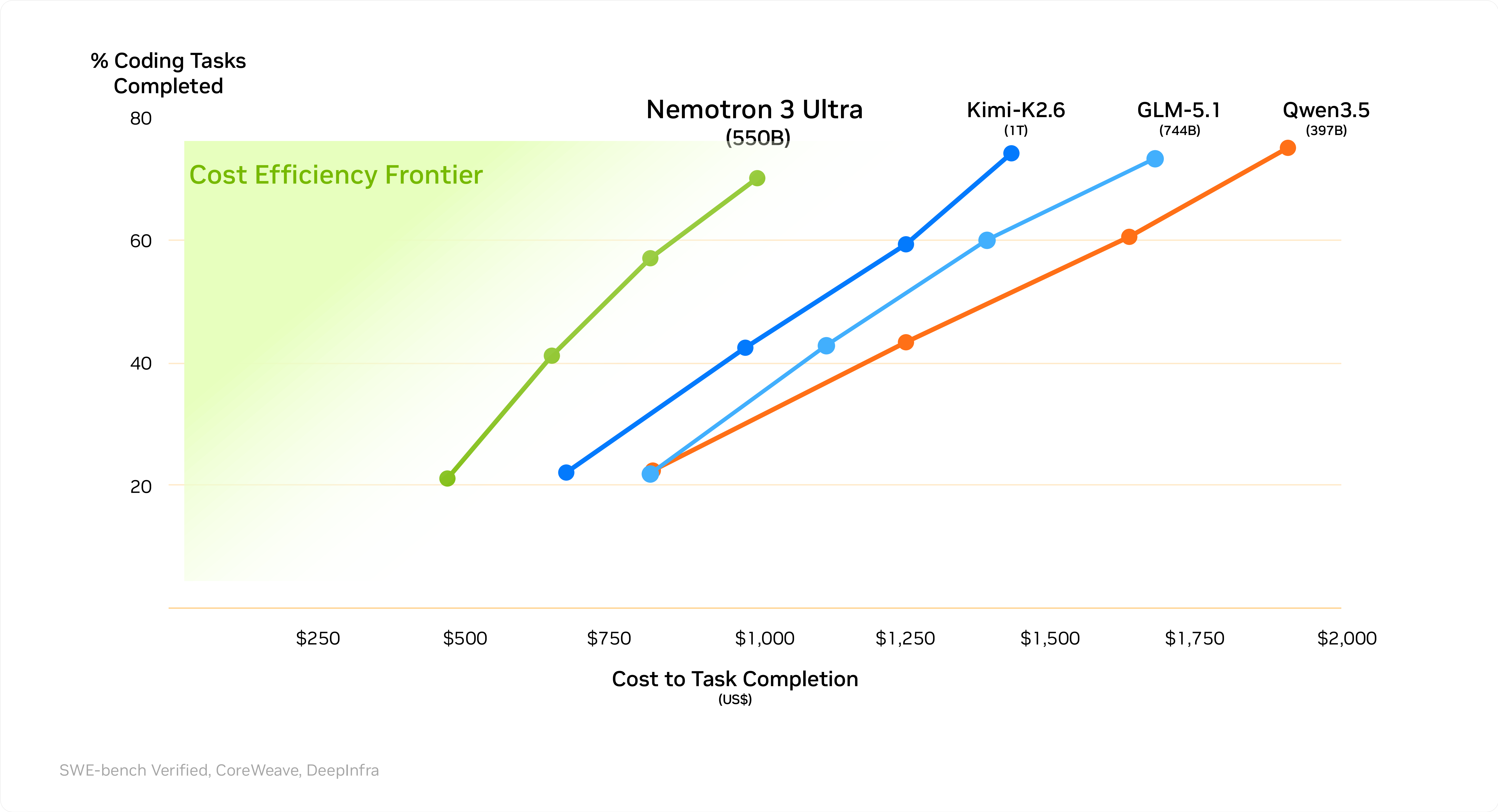
Artificial Analysis Intelligence Index · Output speed: output tokens per second

Most attractive quadrant

Alibaba Kimi NVIDIA Z AI

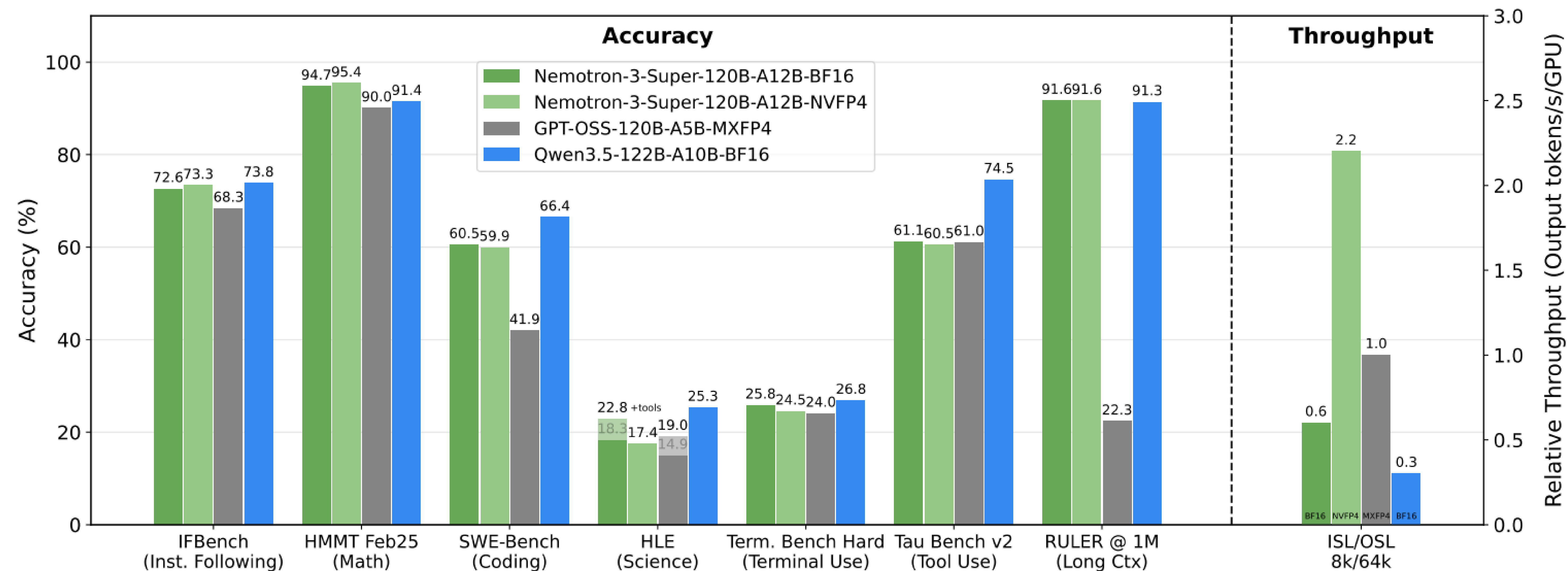


Nemotron 3 Ultra Saves Up to 30%

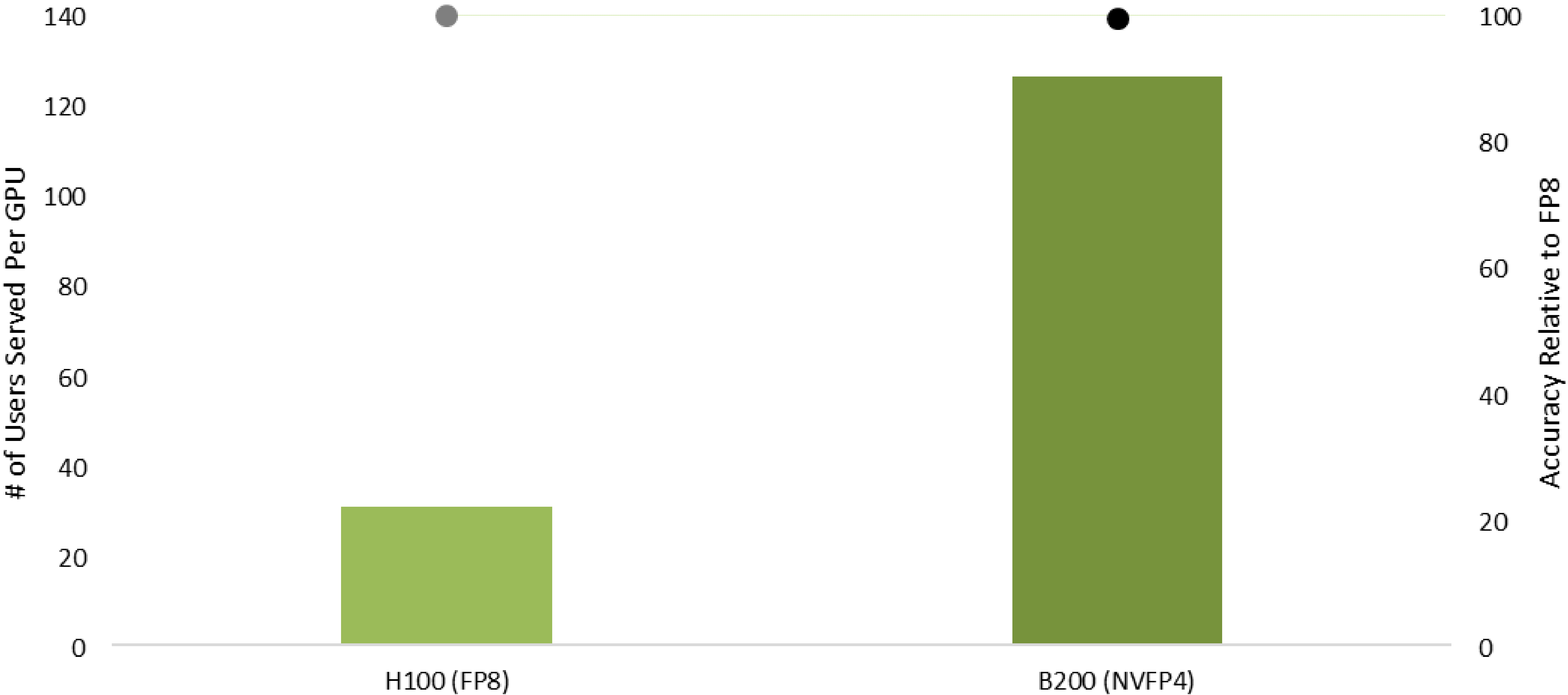


Highest Accuracy for Enterprise Agents Making Complex Reasoning Decisions

Maximize Token Generation and Minimize Inference Cost

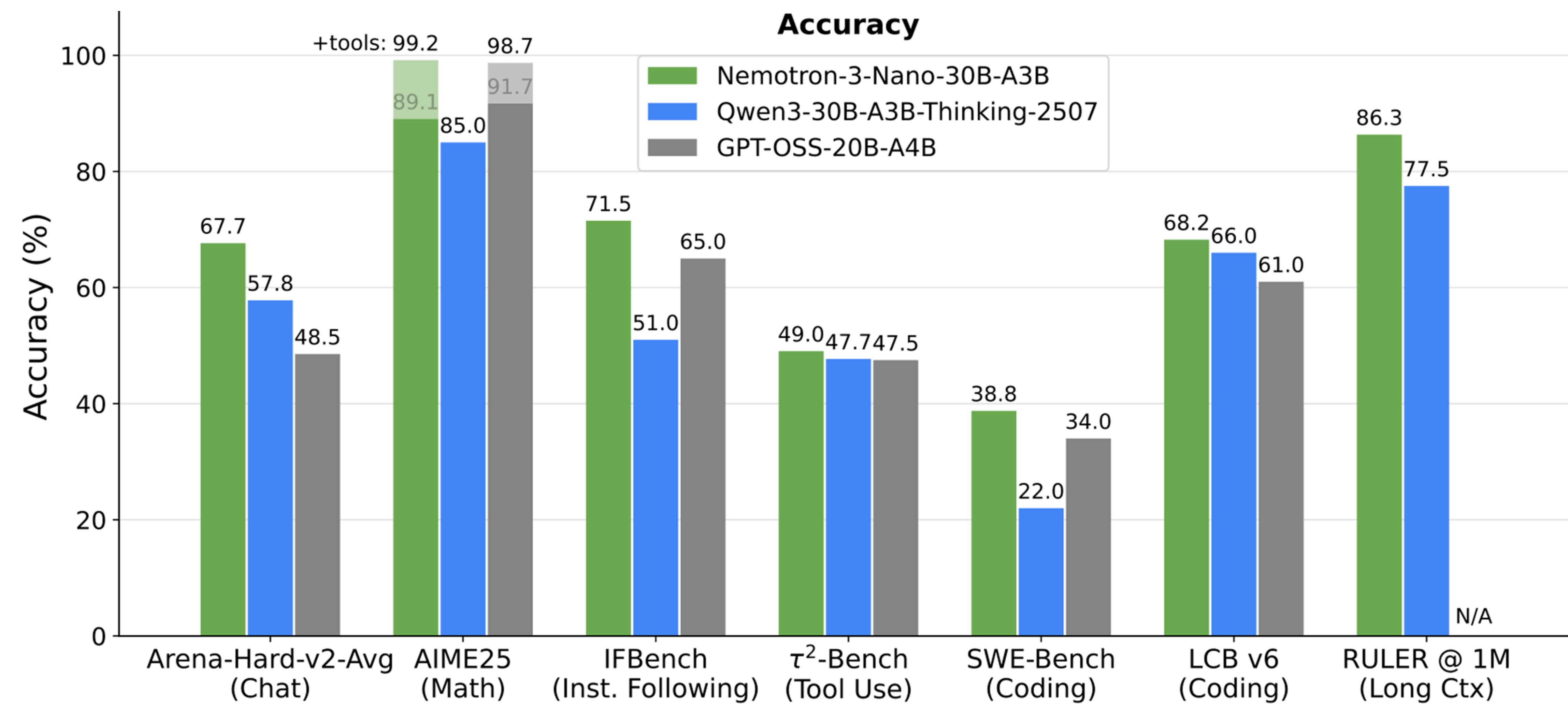


Nemotron 3 Super Achieves 4x Throughput While Maintaining Accuracy w/ NVFP4



Note: 128 Batch Size; Context 8k/64k; 8x NVIDIA B200 and H100; 40 TPS/User

Power AI Agents with Advanced Reasoning Using NVIDIA Nemotron 3 Nano



Coding
SWE Bench, LCB



Scientific Reasoning
GPQA Diamond



Complex Math
AIME 2025



Long-Context
Ruler



Instruction Following
IFBench



Tau2
Tool Calling



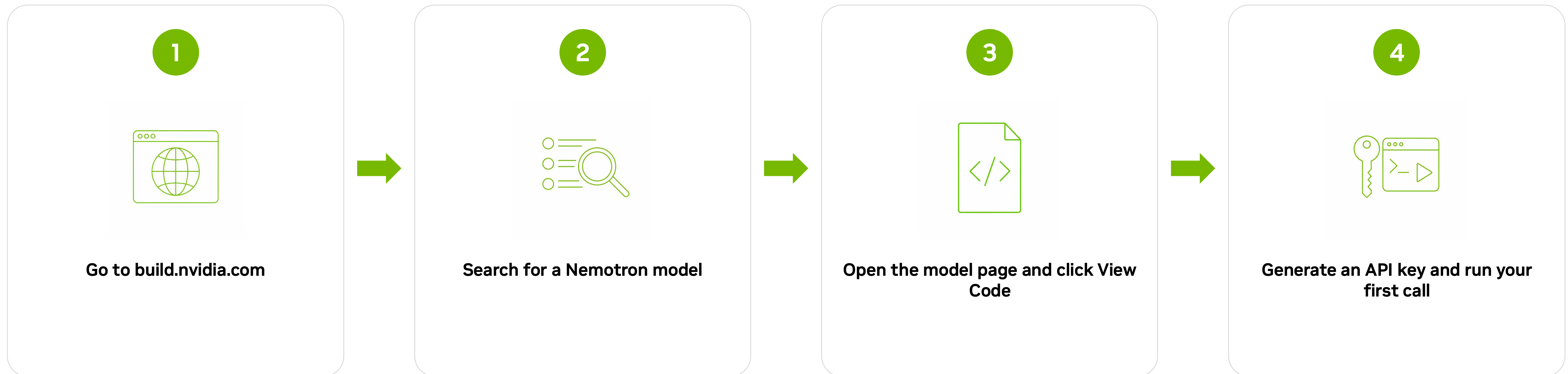
Chat
Arena Hard

Accuracy average of: GPQA-D, AIME25, Ruler@256k, Ruler@1M, IFBench, Arena Hard v2, SWE Bench, Live Code Bench, Terminal Bench, Tau 2 (Telecom, Retail, Airline Average)

If You Take One Action After This Talk, Start Here

Try NVIDIA Nemotron on build.nvidia.com

<https://build.nvidia.com/models>



Hosted API access with sample code on the model page

Open models also available on [Hugging Face](#)

build.nvidia.com → Nemotron → View Code → Generate API Key



Thank you!